NFDI4Objects Community Cluster (CC) Objects as Information Carriers

Christoph Klose (SMB-SPK, Münzkabinett Berlin)

Chairs: Elisa Roßberger (FU Berlin),
Bernhard Weisser (SMB-SPK, Münzkabinett Berlin)

NFDI4 **Objects**

The CC within NFDI4Objects

NFDI4Objects is an initiative within the German National Research Data Infrastructure (NFDI) which pursues the establishment of a sustainable research data infrastructure on a national and international level according to the FAIR and CARE principles. NFDI4Objects represents a broad range of disciplines from the humanities, cultural studies and natural sciences with an archaeological and historical focus dealing with the material legacies of past human activities: Artefacts, architecture, archaeological features, anthropogenically shaped landscapes and ecological remains [1]. Community Clusters are devised according to mutual research data management (RDM) needs and made up of individuals from the NFDI4Objects member institutions, specialists from outside and all other interested persons. The Cluster Objects as Information Carriers aims to establish an open community representing both collecting and editing institutions as well as researchers dealing with text- and image-bearing artefacts from different regions and periods.

Main Objects of the Cluster

The key focus of the cluster is the pursuit of an adequate modelling for the research data of both semantic facets of written objects: information from text, letters, signs and from the material design of objects as well as their precise relationship. Key factors for this type of modelling were identified and solutions discussed during workshops in October 2024 and April 2025 [2]. These encompass above all the precise location of written features on the objects, the relationship of the textual information in comparison to the object it is written on, its design or iconography, or external temporal and spatial entities.

Concept Definition in LOD Thesauri

The medallion of the artist Leone Leoni designed in honour of the renaissance master Michelangelo c. 1560 (**fig. 1 & 2**) serves as a case study to demonstrate the strengths of authority data–based object data capture using the data model of the Digital Coin Collection Portal developed by the Coin Cabinet of the Berlin State Museums (IKMK.net), while simultaneously highlighting shortcomings and desiderata in the current Linked Open Data (LOD) landscape.
The obverse (**fig. 1**) of the famous gilded bronze medallion depicts the elderly universal artist Michelangelo Buonarotti as draped bust to the right, identified by an **inward-facing inscription** with **ligatured lettering**. The inscription conveys additive information regarding the age of Michelangelo at the time the medal was designed and thus providing a clue to dating [3]. At the arm section of the bust the medalist has placed his **signature**, which reads **LEO** and is **oriented outwards**.
While **numismatic standard descriptive information** such as the material of the medallion (bronze) and the techniques of its manufacture (cast and gilded) are expressed **via standardized LOD-vocabularies** (e.g. NDP, Nomisma, Getty AAT), there are **shortcomings in the capture of its written content**. We have segmented various sections of the medallion's inscriptions to highlight this point.
Most important, there is a lack of precision in the definition of the written content, which is either solely designated as (coin) **legend** or missing even in special vocabularies. This ignores an important distinction between **different types of written content** on coins and medallions, which in this case is illustrated by the signature of the artist LEO on the arm of the bust. The **concept of signature** is neither yet separated from the general legend, nor it is thoroughly represented in LOD-vocabularies. The same is valid for other types of inscriptions, including: mint marks, titles or quotations as exemplified on the obverse (see below).
The precise rendering of the inscription's characters (**lettering**) is likewise not fully captured. For instance, the data model does not record the **direction** of the characters, neither whether the letters are executed in relief or recessed, nor whether they appear in uppercase or are joined as **ligatures**, although corresponding concepts exist in LOD vocabularies.

Fig. 1: Obverse of Leone Leoni's bronze medallion (MK Berlin 18210913) https://ikmk.smb.museum/object?id=18210913 with segments of its inscribed parts and relating or missing concepts in LOD-Vocabulary

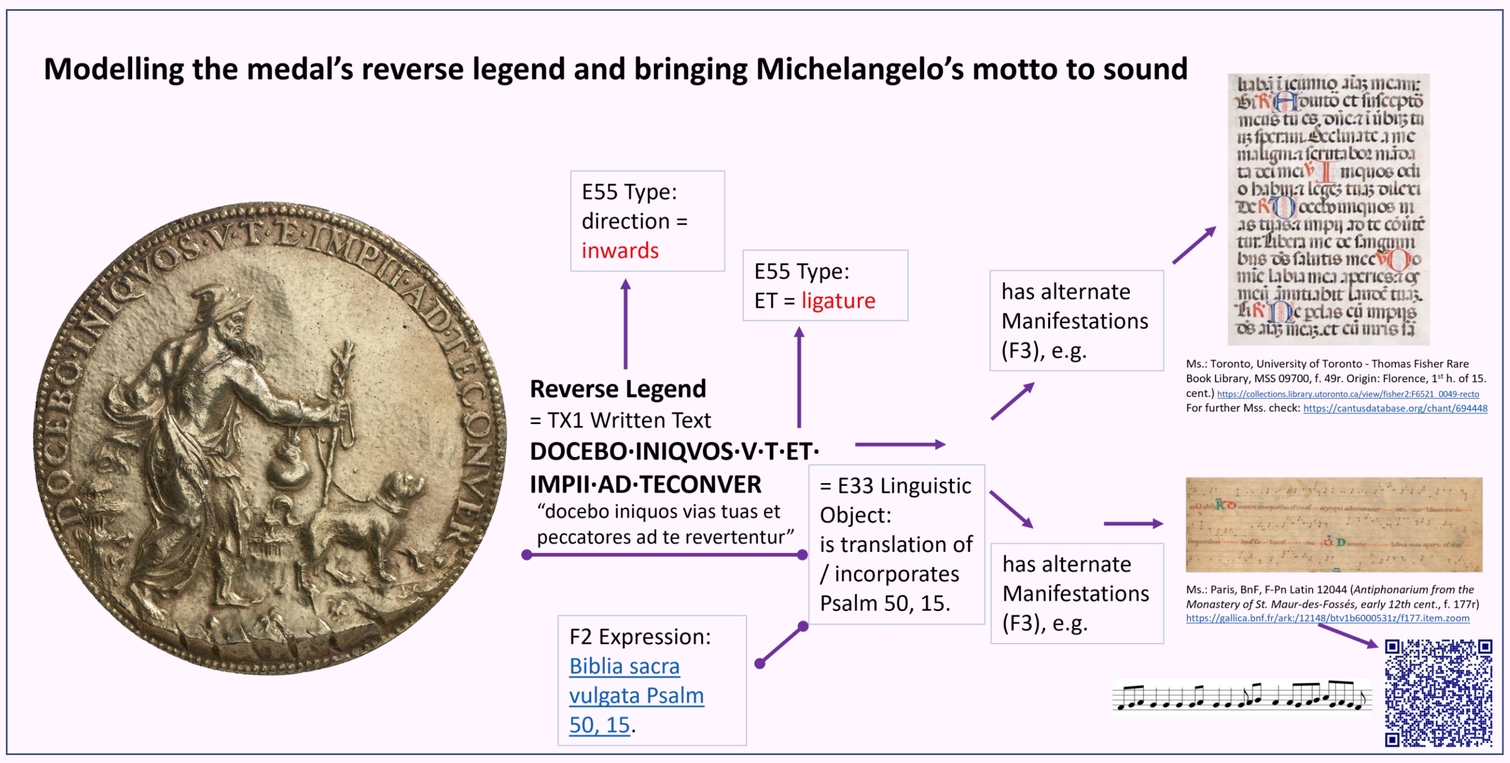

Fig. 2: Reverse of Leone Leoni's bronze medallion (MK Berlin 18210913) https://ikmk.smb.museum/object?id=18210913 with a modelling of the reverse legend as a quotation of Psalm 50 and as edited in manuscripts and in medieval musical notation – according to the FAIR ontology model

How to make a medal ‚sing'

It may also be of interest to more precisely define the nature of written information in relation to its original source. This can also be the case for numismatic objects, which is why, in **fig. 2**, we attempt to illustrate the relationship between the motto chosen as reverse legend of the medallion, its function and multimodal tradition by testing some key properties of the **CIDOC CRM based FAIR Ontology model** [4]. In this way, we also aim to illustrate how we can make the medallion's inscription 'sing'.
Both the text and the image on the reverse of the medallion have elicited scholarly debate [3]. The textual component, which is the primary focus of this analysis, is an abbreviated version of Psalm 50,15 from the Latin Vulgate. This specific selection from the Psalms holds particular significance in Christian liturgy and theology, often associated with themes of repentance, sacrifice, and divine invocation. Interestingly, the old pilgrim being led by a dog along a stony path seems to best illustrate the most abbreviated part of the motto V(ias) T(vas), which suggests a very precise coordination between image and text. The rich reverse side depiction of the medal, which has given rise to a variety of interpretations, is, deliberately contrasted with the brevity of the textual description.

In **CIDOC terms**, the reverse legend is the **written text (TX1)** of an abridged version of the **linguistic object (E33)** containing the **expression (F2)** from Psalm 50:15. The linguistic object also has **alternative manifestations (F3)**, which can include both textual forms (e.g. a 15th-century manuscript from Florence) and musical annotations for liturgical purposes, such as those used on Passion Sunday (in a 12th-century manuscript now located in Paris). A (rough) audio reproduction of this melody can be downloaded via the QR code. This brings us closer to Michelangelo's motto, which may also have been chosen with a melody in mind.

Join our Community

We appreciate all types of co-operation, partnership and support. New enthusiastic members are always welcome!

Mailinglist

Rocket Chat Channel

SPK berlin-brandenburgische AKADEMIE DER WISSENSCHAFTEN

EPIGRAPHY info

Epigraphy.info Meeting X in Graz 24.-26. March 2026

References

[1] D. Bibby, K.-C. Bruhn, F. Dührkohp, et al. (2021). Digitales Forschungsdatenmanagement in der Archäologie und die Initiative NFDI4Objects, Blickpunkt Archäologie, vol. 2, pp. 150–163, 2021. https://doi.org/10.5281/zenodo.5823868; Bibby, D., Bruhn, K.-C., Busch, A. W. et al. NFDI4Objects - Proposal. https://doi.org/10.5281/zenodo.10409228

[2] CC Objects as Information Carriers: Online-Workshop 22.10.2024, Die Materialität von beschriftetem Kulturgut im digitalen Raum: https://osf.io/hqtcx/files/osfstorage; Objekt – Text –Bild, Workshop Berlin, Brugsch-Pascha-Saal, 28.4.2025: https://doi.org/10.5281/zenodo.17950254

[3] For full discussion cf. A. Schumacher, Leone Leonis Michelangelo-Medaille, in: G. Satzinger (ed.), Die Renaissance-Medaille in Italien und Deutschland (Münster 2004) 169-194

[4] J. Prag, I. Asif, P. Hermankova, M. Horster, Building an epigraphic ontology (2025) https://epigraphy.info/workshop_9_posters/#poster13; https://ontology.inscriptiones.org/epont